

How GPS Works

Triangulation Game | Grade 3 AI & Digital Literacy

In this lesson you will discover how GPS (Global Positioning System) uses satellites in space to find exactly where you are on Earth. You will also play the Triangulation Game to experience how it works yourself!

Learning Goals

- Understand what GPS is and why we use it.
- Learn how satellites send signals to find your location.
- Discover what triangulation means — step by step.
- Play the Triangulation Game with your classmates!

What is GPS?

GPS stands for **Global Positioning System**. It is a system that helps us find places anywhere in the world — like a magic map inside your phone!

GPS works by talking to special machines called **satellites** that fly high above the Earth. These satellites send invisible signals down to your phone or GPS device, and the device uses those signals to figure out exactly where you are.

GPS is super helpful every day — for example:

Finding your way
home

Tracking school buses
or cars

Playing location games

Meet the Satellites!

GPS satellites are like **helpful friends in space**. They fly around the Earth in a big orbit and send special signals down to us all the time.

There are **more than 30 GPS satellites** circling the Earth right now. They work together in teams to help pinpoint your exact location on the ground.

Three Key Satellites Used for Triangulation

Satellite 1 — sends the first signal that creates a big invisible circle around you.

Satellite 2 — sends a second signal; where the two circles cross narrows down your area.

Satellite 3 — sends a third signal; where all three circles meet is your EXACT spot!

What is Triangulation?

Triangulation is the key to GPS magic! It is the method of using signals from **3 satellites working together** to pinpoint your exact spot on Earth.

How It Works — Step by Step

Step 1	Step 2	Step 3
Satellite 1 sends a signal. Imagine a big invisible circle spreading out from it. You are somewhere on that circle.	Satellite 2 sends a signal too. Now we have two circles. Where they cross narrows your location to just a small area.	Satellite 3 sends a third signal. All three circles meet at ONE point — that is exactly where you are standing!

3 satellites working together can pinpoint your spot!

GPS Around Us

GPS Helps Everyone, Everywhere!

GPS is used by many people every day to find their way and stay connected with the world around them.

Navigation Apps	Tracking Vehicles	Pilots and Ships
Help us find routes to new places.	Know where buses and cars are.	Safely travel through sky and sea.

Why is GPS Important for You?

GPS Benefits for Kids

Keeps you safe: GPS helps your family find you.

Makes exploring fun: Discover new places with maps!

Connects us to the world: GPS links everyone everywhere!

Triangulation Game Instructions

Now it is your turn to try the triangulation game! Grab some friends and strings and head outside.

1	2	3	4
Use 3 friends as "satellites" standing apart.	Each friend stands at a different point.	Use string or chalk to make circles around each friend.	Find where all circles overlap — that is your spot!

Quick Quiz: What Did We Learn?

Test your GPS knowledge! Write your answers below.

No.	Question	Your Answer
1.	What does GPS help us do?	
2.	How many satellites does GPS use for triangulation?	
3.	What is triangulation?	
4.	Name one way GPS is used in everyday life.	

Fun Fact: GPS Satellites

There are more than **30 GPS satellites** circling Earth right now! They travel about 20,200 km above us and complete two full orbits of the Earth every day. They work together to help us every single day.

What We Learned

Term	Meaning
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GPS	A system that uses satellites to find your location anywhere on Earth.
Satellite	A machine in space that sends signals down to Earth.
Signal	An invisible message sent from the satellite to your device.
Triangulation	Using 3 satellites together to find your exact spot.

Thank you for learning how GPS works! Keep playing, exploring, and discovering!